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Distance-based confidence generation and aggregation of classifier for unstructured road detection

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ABSTRACT

Road region and non-road region separation in the unstructured road intends to be an important task for safe navigation and collision avoidance for autonomous driving vehicles. The road that connects rural areas and cities to the national highways are considered as unstructured roads. Absence of clear lane marking on these unstructured roads makes them more prone to accidents in comparison to highways, which have clear lane marking for indication of road and non-road regions. However, the unstructured roads have different color information from its background that paves an easy way for design and development of an efficient road detection system for recognition and classification of road and non-road regions. Hence, in this paper, we propose an efficient road detection system for the classification of unstructured roads into road and non-road regions using multiple nearest neighbors (NN) classifier and soft voting aggregation approach. The proposed system utilized the chromatic information (i.e. *a, *b, and Hue) to train the NN classifiers and aggregated their output using soft voting (SV) approach for final output response. The output results of multiple classifiers were aggregated using SV approach based on the confidence score obtained by each individual classifier.

The performance of the proposed system is evaluated in terms of precision, recall, accuracy, intersection over union (IOU), true positive rate (TPR), and processing time and compared with current state of art methods reported in the literature. The proposed system achieved precision, recall, accuracy, IOU, and TPR of 96.79%, 96.92%, 97.8%, 96.08% and 96%, respectively with the processing time three times smaller than those of the existing state of art methods. The experimental results demonstrate that the proposed system can provide an effective guidance to the autonomous vehicles through recognition and classification of road and non-road regions in rural, urban, and city areas, wherein single unstructured roads connect the national highways.

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1. Introduction

Accurate path detection on structured and unstructured roads plays a key role in the safe navigation of autonomous vehicles. Various sensors that have been used in autonomous vehicles play a critical role in the accurate detection of path regions on structured and unstructured roads. Based on the utilized sensor in path detection, road detection problems can be classified into two classes such as camera-based detection (Jiang et al., 2013; Kim, 2008) and light detection and ranging (LIDAR) based detection (Wijesoma et al., 2004; Landa and Prochazka, 2014). LIDAR-based autonomous vehicles are very expensive due to the high cost of the LIDAR sensors. Therefore, nowadays, vision-based road detection systems are becoming more popular due to their low cost, rich content information, and faster adaptability towards emerging technologies such as machine learning and the internet of things (IoT).

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Based on the configuration of roads, they can be categorized into two classes such as; structured and unstructured roads. The roads which have clear lane marking like the highways are considered structured roads, while the roads that do not have any clear lane markings are considered non-structured roads. Over the last ten years, numerous methods have been proposed by researchers for the detection of structured roads and achieved satisfactory results (Thorpe, 1993; Li et al., 2004; Wang et al., 2004). Therefore, the structured path detection problem is considered a solved problem in the current scenario (Caltagirane et al., 2017). Although many experiments have been conducted for unstructured road detection (Rasmussen, 2012; Wang et al., 2009; Audibert et al., 2010; Cheng et al., 2006) too, it requires more attention due to its complex, variable, and unstructured nature. In literature (Jiang et al., 2013; Al-Jarrah et al., 2018; Hummel et al., 2006), features namely color, edges, and textures have been used frequently for detecting unstructured roads. However, the selection of color space is crucial and plays an important role in image segmentation and computer vision tasks. In literature, Hue Saturation Value (HSV) and L*a*b color space have been employed successfully by eminent researchers for different applications such as medical image segmentation (Bock et al., 2010; Harrabi and Ben Braiek, 2012; Chevrefils et al., 2009), fruits identification (Li et al., 2014; Unay and Gosselin, 2007; Vélez-Rivera, 2014), human segmentation (Borràs and Lladós, 2009), sea-land segmentation (Shui et al., 2020), object detection (Chen et al., 2015) and traffic sign detection (Ruta et al., 2010) due to their inherent capability of extracting robust features even in the dynamic environment. These features have been proven useful in light varying conditions too (Wu and Sun, 2013; López-Rubio et al., 2016). In various research assignments, the chromatic information has been utilized successfully for pattern recognition tasks to train the different machine learning models such as support vector machine (SVM) (Wang et al., 2009), nearest neighbor (NN) (Recognition, 2011), K-nearest neighbor (KNN) (Chevrefils et al., 2009), artificial neural network (ANN) (Chen et al., 2015), and decision tree (DT) (Ruta et al., 2010) and achieved the promising results using these algorithms.

Such successful applications of machine learning algorithms with chromatic features motivated us to adopt these features for the detection and classification of unstructured roads. Hence, in this study, we utilized chromatic feature (i.e., *a, *b, and Hue) information as an input to build the NN classifier. The proposed work is an important application from the perspective of the developing countries like India that has the huge potential to become the largest market for autonomous vehicles in the forthcoming years.

1.1. Related work

In this section, we present various methods and techniques, which are available in literature for the detection of unstructured roads and similar tasks. For instance, Audibert et al., (Audibert et al., 2010) proposed a method based on vanishing point and edge detection for the identification of road boundaries. In their study, the authors used the 5 different frequencies scale and 36 different texture orientations to detect the vanishing point. The vanishing point represents a sharp boundary point, where the road region ends and the non-road region starts. Therefore, during the detection of the road regions in unstructured roads, one needs to discern the vanishing point. The proposed method achieved satisfactory results; however, the major limitation of this method is that it relies on the high number of image planes (180 image planes), which enhance the system complexity and make it economically non-affordable. Jinjin Shi et al., (Shi et al., 2016) presented a less complex vanishing point detection method to further reduce the system complexity in which only the dominant texture features and particle filter's output features were analyzed for the detection of vanishing points. Moreover, Tan et al. (Tan et al., 2006) formulated mathematical modeling for the detection of road boundary of the unstructured roads by calculating its related parameters. They used multiple frame sequences and color information to distinguish between the road and non-road regions.

In recent years, researchers have made sincere efforts to enhance the capability of unstructured road detection systems by integrating different types of sensors with vision sensors. For instance, Sonet et al. (Son et al., 2008) combined the posterior probability of the laser sensor with the visual information of the camera to detect the unstructured roads. Shang et al. (Shang et al., 2013) combined GPS, camera, and LIDAR sensors to detect the road regions on the unstructured roads and subsequently used various machine learning algorithms such as support vector machine (SVM), and k-nearest neighbor (KNN) for the classification of the region of objects. Nowadays, deep learning techniques such as convolution neural networks (CNNs) are becoming very popular and have been proven efficient for pattern recognition, object detection, and vision application tasks. However, deep learning has not been used at a large scale as of now for the detection of unstructured road detection, owing to the non-availability of a significant amount of relevant data to train the models. In Ref. (Caltagirane et al., 2017), authors employed convolution neural networks for the detection of the drivable road regions. Besides, Pan et al. (Pan et al., 2017) suggested a hybrid approach based on Markov Random Field (MRF) and CNN to detect and classify the traffic scenes on roads.

Despite offering various advantages, deep learning algorithms offer several limitations. For instance, deep learning (DL) models are computationally expensive due to the usage of many hidden layers (Lu et al., 2017; Cao, 2019), which need to fine-tune the hyper-parameters to yield the desired result. Apart from this, deep learning algorithms require large volume of data, an outsized memory device, more processing time, and high-performance hardware such as graphics processing units (GPU) for developing an efficient road detector system (Aziz et al., 2020; Zhao et al., 2020). Whereas the lane detection systems or road region detectors need to be computationally fast and less complex to take a quick decision (Zuo et al., 2019; Patle et al., 2019). Hence, this research work focuses on developing a simple and effective road detection and classification system to effectively differentiate between the road and non-road regions in unstructured road images.

In this work, we used the nearest neighbor classifier and soft voting approach to develop an efficient unstructured road detection method. As stated above, color features might provide crucial information about road and non-road regions. Therefore, in the proposed study, we used color information as the input features to build the classifier. The extracted color features were fed as input to the three independent nearest neighbor classifiers. Subsequently, the confidence score (CS) of every classifier was generated based on the minimum distance between the training sample and the sample under study. Finally, it got aggregated by using the soft voting aggregation approach to yield the final output.

The contributions of the present study are:

(1) To develop an efficient and novel road detection system for the classification of unstructured roads into the road and non-road regions using multiple nearest neighbor (NN) classifier and soft voting aggregation approach.

(2) To analyze the performance of the proposed system using different performance metrics such as precision, recall, and accuracy as well as in terms of precision vs. recall receiver operating characteristics (ROC) curve.

(3) To compare the performance of the proposed method with other conventional machine learning and deep learning models.

The rest of the paper is organized as follows: a detailed description of the image acquisition, image pre-processing, features

extraction, and development of NN classifiers are covered in section 2. The obtained results and simulation details are described in section 3. Finally, the proposed work is concluded in section 4.

2. Proposed road detection system

Fig. 1 shows a basic flow diagram of the proposed road detection system. The proposed system includes four major steps: (i) Image acquisition, (ii) color feature extraction, (iii) building the classifier, and (iv) aggregation of confidence score (CS) using the soft voting approaches to yield the final output. In the following section, each step of the proposed approach is described in considerable detail one by one:

2.1. Image acquisition

To develop the proposed system, 4000 sample images of unstructured roads were collected from Faridabad and nearby region of Delhi, India using a digital camera of resolution1600x1200 pixels. The output performance of vision-based road detection systems mainly depends upon the illumination quality of the captured images. Therefore, all the images were captured during daytime from 6 A.M in the morning to 6P.M in the evening. Due to the acquisition of the images at different timings and under different lightning conditions, the collected images had adequate illumination variation that makes them suitable for developing a robust road detection system. While capturing the images, the latitudes of the captured image region were fixed between 28°-24′-17″ and 28°-53′-00″ North, while the longitudes were kept between 76°-50′-24″ and 77°-20′-37″ East. All the captured images exhibit RGB color space. Fig. 2 presents some of the sample images which were acquired during daytime under different light conditions. The generated data set includes the images of the road and non-road regions in almost equal numbers to avoid the over fitting problem of the classifier. The individual set of sample images representing road and non-road regions are illustrated in Fig. 3.

2.2. Image pre-processing

Pre-processing is considered an essential step in the field of image processing, which is generally performed to improve the image quality through suppression of redundant information as well as enhancing the object features in an original image. In pre-processing, the input image is made suitable for subsequent processing by passing it to the number of stages such as image cropping, image resizing, image denoising, and image sharpening through the application of appropriate filters. Initially, road and non-road regions from original images were manually cropped into the size of 228 × 228 pixels to reduce the computational burden on the hardware, which, in turn, ultimately helps in improving and escalating the decision process of the classifier (Gao and Wang, 2007; Murty and Devi, 2011; Kozakiewicz, 2018).

After the image cropping, image resizing, image de-noising, and image sharpening algorithms are applied to the resulting images to highlight the desired feature information.

2.2.1. Image down-sampling and smoothening

In this work, a high-resolution camera was used for image acquisition. The high-resolution image needs large memory and thereby lays more computational burden on the machine. Therefore, to address the issue, the size of cropped images was further reduced to 50% of their actual size through the process of down sampling. A pilot experiment was conducted to observe the effect of down-sampling on the speed performance of the system and it was found that the speed performance of the system improved significantly after down sampling. Therefore, in this work, all the images were down-sampled before applying them as an input to the classifier. To down-sample the image, every image pixel value was replaced by weighted pixel values of its neighborhood using a Gaussian filter mask of size 5x5 pixels. Gaussian filter mask used the standard deviation of 1 for replacing the information with its neighborhood (Sokolov and Abdullin, 2017). Adoption of this process suppressed the background information to some extent. The Gaussian filter can be mathematically expressed using (1). Eqs. (2) Indicates the operation of the Gaussian filter on the cropped image to yield the down sampled image.

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\left(\frac{x^2+y^2}{2\sigma^2}\right)} \tag{1}$$

$$g(x,y) = \sum_{(i,j=1)}^{m} w(i,j) * f(x+i, y+j) \tag{2}$$

where *f(x,y)* is the captured image, *m* represents the size of the filter mask, *w* shows the mask of filter, and *g(x,y)* represents the image obtained after applying the Gaussian filter. Gaussian filtered image is then stored in the form of a new matrix, which is just half of the size of the original image matrix. The mathematical expression for image resizing is given by Eq. (3).

$$r\left(\frac{x}{2}, \frac{y}{2}\right) = g(1:2:x, 1:2:y) \tag{3}$$

where, *g(x,y)* is a Gaussian filtered image, *x* and *y* is the original image size, and *r(x,y)* is the resized image.

The down-sampling process highly suppressed the image voxels to yield the image of the desired size, which eventually impacted the image quality. Owning to which, the edge features of the foregrond region got blurred. Therefore, in order to restore the suppressed pixels of especially foreground region, we again applied the gaussian filter of size 3x3 pixels with a standard deviation of 0.5. Thus, the value of suppressed pixels got restored by convolving the pixels of the resized image with the Gaussian filter, resulting into an image that was more suitable for subsequent processing as illustrated in Fig. 4 (Row 2). The mathematical equation for the enhanced image is represented using Eqs. (4).

$$En(x,y) = \sum_{(i,j=1)}^{m} w(i,j) * r(x+i, y+j) \tag{4}$$

Where *w* is a weighted Gaussian mask, *r(x,y)* is a resized image and *En(x,y)* is the output de-noised image.

2.2.2. Image sharpening

Image sharpening is used to highlight the edges and fine details in an image. In this study, we used a high pass filter to sharpen the image features. The high pass filter extracted the high frequency information as shown in Fig. 4 (Row 3), which represents the edges and fine details of foreground regions. Thereafter, the resulting high frequency image was superimposed on the enhanced image to produce the final sharpened image (Polesel et al., 2000). The resulting final sharpened image is depicted in Fig. 4 (Row 4). Mathematical expression for image sharpening is given by Eq. (5).

$$s(x,y) = En(x,y) + \lambda * h(En(x,y)) \tag{5}$$

where, λ is the scale factor,$h(En(x,y))$ indicates high pass filtered enhanced image and $En(x,y)$ is the enhanced image. s(x, y) is reffered as sharpened image.

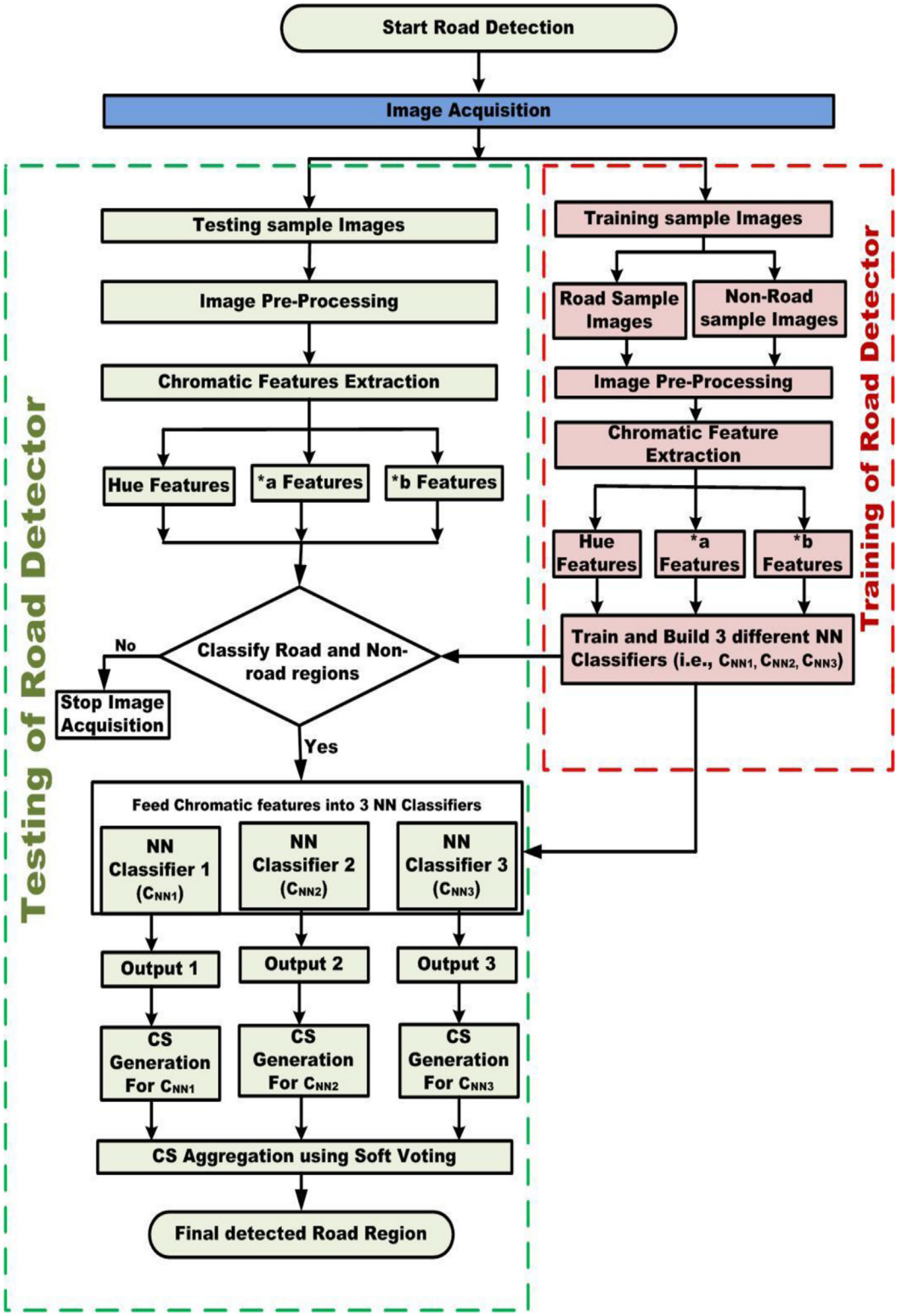

Fig. 1. Process flow diagram of the proposed road detection system.

2.3. Feature extraction

Features represent specific information about the road and non-road region. It helps the classifier to learn about a pattern of objects and in the decision-making process. Hence, feature extraction is an important step for building an efficient classifier. Connecting streets between highway and city have different color information for road and non-road (i.e. background) regions. Hence, color information is considered critical for the detection and classification of unstructured roads. As stated above, RGB color space information highly depends upon the variations in light conditions, in which the image is captured and tends to affect the classifier performance

Fig. 2. Samples of the acquired images under varying light conditions.

(a) (b)

Fig. 3. Training sample images: (a) Sample images of road region, (b) Sample images of non-road region.

if RGB color space is directly used as feature. Therefore, we used chromatic information, which was derived from other color spaces such as L*a*b color space and HSV color space. Chromatic and hue information has been extracted from CIE L*a*b* color space and CIELCh color space, respectively. Chromatic information *a, *b is chosen from CIE L*a*b color space, while hue angle (H) information is extracted from CIELCh (Paschos, 2001; Schloss et al., 2018). Fig. 4 (Row 5–7) shows the extracted image plane of chromatic *a, chromatic *b, and extracted image plane of Hue information, respectively.

The transformation of RGB to L*a*b color space was carried out using the procedure given in (León et al., 2006). The chromatic information such as *a,*b, and Hue angle, H can be calculated using (6), (7), and (8) as follows:

$$a^* = 116\left(f\left(\frac{X}{X_n}\right) - \left(\frac{Y}{Y_n}\right)\right) \tag{6}$$

$$b^* = 116\left(f\left(\frac{Y}{Y_n}\right) - \left(\frac{Z}{Z_n}\right)\right) \tag{7}$$

$$H = \tan^{-1}\frac{*b}{*a} \tag{8}$$

where, X_n, Y_n, and Z_n are the CIE XYZ tristimulus values of the reference white point.

The average value of each transformed sample image is calculated and stored as a feature vector, *Av*. The feature vector can be mathematically expressed as:

$$A_V = \frac{1}{M * N}\sum_{i=1}^{M}\sum_{j=1}^{N} T(i,j) \tag{9}$$

where, *Av* represents the mean value, *(M, N)* is the number of rows and number columns of the sample image.

T (i, j) represents the sample image of the region.

The feature vector of the road (1500 samples) and non-road regions (1500 samples) are calculated by taking the mean of feature vectors, which is computed using Eq. (9). The mean values of the feature vectors can be mathematically computed as:

$$F_{VR} = \frac{1}{1500}\sum_{i=1}^{n} A_{VR} \tag{10}$$

$$F_{VNR} = \frac{1}{1500}\sum_{i=1}^{n} A_{VNR} \tag{11}$$

The F_{VR} and F_{VNR} are the final feature vectors, which are formed from the feature vectors pertaining to different color components of chromatic and Hue information and can be expressed as$F_{VR} = [F_{VR*a}, F_{VR*b}, F_{VRH}]$ and $F_{VNR} = [F_{VNR*a}, F_{VNR*b}, F_{VNRH}]$, for road and non-road regions.F_{VR} and F_{VNR} represent the mean value of feature vectors for road and non-road regions, respectively.

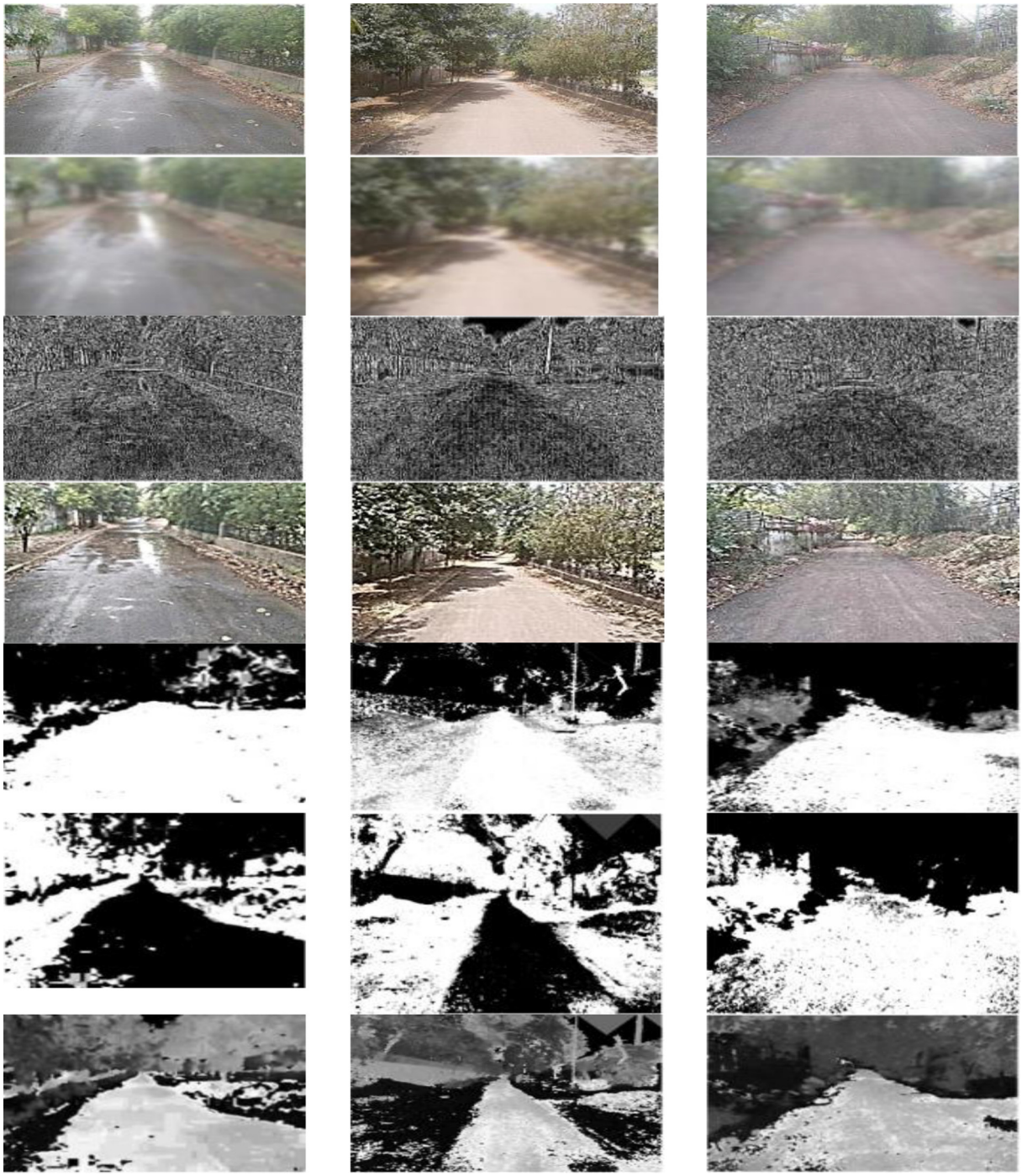

Fig. 4. Row 1: Original images after cropping; Row 2: Image after down sampling (i.e. resizing) and applying Gaussian smoothening filter; Row 3: Images after applying High-frequency filter; Row 4: Sharpened image samples; Row 5: Transformation of RGB image into Chromatic *a feature; Row 6: Transformation of RGB image into Chromatic *b feature; Row 7: Transformation of RGB image into Hue feature.

The extracted chromatic feature information after being transformed into a feature vector set was used as an input to train the classifier. The development phase of the classifier is discussed in detail in the following section.

2.4. Building the classifier

In this work, we used three independent nearest neighbor classifiers, which are denoted as C_{NN1}, C_{NN2}, and C_{NN3}. We trained these three independent classifiers with three independent features viz., *a,*b, and H. Each classifier generated a specific confidence score based on the calculation of the minimum distance between training feature vector points and test feature vector points. The outputs of all these classifiers were aggregated using a soft voting approach based on the maximum vote score for making the final decision.

The proposed road detection system, which comprises the nearest neighbor algorithm and soft voting approach, classifies the unstructured road images into the road and non-road regions based on the least distance from a particular class (Gao and

Wang, 2007). The methodology for feature extraction and feature vector formation has already been described in the previous section and therefore we can directly denote the feature vectors as (F_{VR}, θ_R) and(F_{VNR}, θ_{NR}), corresponding to road and non-road regions, where F_{VR} and F_{VNR} represents features vectors and θ_R and θ_{NR}indicates labels for road and non-road regions.

The basic operation of the NN classifier can be understood more clearly by the following illustration. Let's p be the feature vector point corresponding to a particular input image and is given to the classifier for labeling or classifying it as a road or non-road regions. As discussed above, the nearest neighbor classifier makes the decision based on the measurement of the least distance between the test feature vector points and training feature vector points from the particular class. Less distance from a particular class indicates more similarity with that class. If the test point is more similar to the road region, then the system assigns 1 to the road region and 0 to the non-road region, otherwise vice versa. The details of the individual classifiers, which are built using independent chromatic (i.e. *a,*b) and Hue (i.e. H) feature information is given as follows:

2.4.1. Building the classifier with chromatic information *a

This section gives an implementation detail of the classifier with chromatic features *a. Initially, in the first step, chromatic feature information *a of both road and non-road regions is fed to the NN classifier as an input in the form of feature vectors, F_{VR*a}andF_{VNR*a} and simultaneously corresponding output labels are assigned to the respective data, which can be denoted as $\theta_R$andθ_{NR} respectively. The classifier model calculates the distance between the test point *p* and the feature vectors of both road and non-road regions and thereby checks for the minimum distance from a particular class (Frank, 2004). Eqs. (12) and (13) show the mathematical representation of distance calculation from the road and non-road regions. Whereas Eqs. (14) Shows the minimum distance selection and Eq. (15) shows the class θ_R orθ_{NR}assigning process.

$$d1 = \|p - F_{VR*a}\| \tag{12}$$

$$d2 = \|p - F_{VNR*a}\| \tag{13}$$

$$C1 = \min\{d1, d2\} \tag{14}$$

$$C1 = \left\{ \begin{array}{ll} \theta_R & \text{if } d1 < d2 \\ \theta_{NR} & \text{otherwise} \end{array} \right\} \tag{15}$$

where, d1 indicates the distance between the test point and road features, while d2 denotes the distance between the test point and non-road features. C1 specifies the outcome of the first classifier with chromatic feature *a as an input.

2.4.2. Building the classifier with chromatic information *b

In building the second classifier, chromatic features information *b is used as an input in the form of features vectors for both types of data (i.e. road and non-road regions). F_{VR*b}andF_{VNR*b} represent the input feature vectors for road and non-road data, respectively, while θ_R*and*θ_{NR} denotes the assigned class labels as road and non-road regions to the training data. The developed classifier calculates the minimum distance between test point *p* and the input feature vectors using Eqs. (16), (17), and (18) and finally yields an output in the form of class labels (i.e. road, θ_R or non-road region, θ_{NR}) as illustrated by Eq. (19). The Eqs. (16–19) can be mathematically defined as:

$$d3 = \|p - F_{VR*b}\| \tag{16}$$

$$d4 = \|p - F_{VNR*b}\| \tag{17}$$

$$C2 = \min\{d3, d4\} \tag{18}$$

$$C2 = \left\{ \begin{array}{ll} \theta_R & \text{if } d3 < d4 \\ \theta_{NR} & \text{otherwise} \end{array} \right\} \tag{19}$$

where d3 is a measured distance between the test point and road features, d4 is a measured distance between the test point and non-road features, and C2 is an output of trained classifier with chromatic feature *b.

2.4.3. Building the classifier with chromatic information Hue

Here, we described the implementation details of third classifier that was trained with Hue feature. To train the classifier, the feature vector sets corresponding to Hue chromatic information was represented as (F_{VRH}, θ_R) and(F_{VNRH}, θ_{NR}), whereF_{VRH} is a feature vector set for road region, while F_{VNRH}depict the feature vector set for non-road region. Similarly, θ_R and θ_{NR} represent the target labels for road and non-road regions, respectively. Likewise two previous classifiers, this classifier also takes into account the minimum distance between the input feature vector and the testing data point *p* using (20), (21) and (22).

$$d5 = \|p - F_{VRH}\| \tag{20}$$

$$d6 = \|p - F_{VNRH}\| \tag{21}$$

$$C3 = \min\{d5, d6\} \tag{22}$$

$$C3 = \left\{ \begin{array}{ll} \theta_R & \text{if } d5 < d6 \\ \theta_{NR} & \text{otherwise} \end{array} \right\} \tag{23}$$

Where, d5 is a measured distance between the test point and Hue features of the road region, while d6 is a measured distance between the feature point of test image and feature points of non-road region image. Whereas, in (23),θ_R and θ_{NR} denotes the target labels as road and non-road regions, with C_3 as a final output of the third classifier.

2.5. Classifiers output aggregation with soft voting approach

As discussed in the above sections, we trained three classifiers, each with independent color feature. We obtained the output of each classifier as a road or non-road region. The final decision on test data was made by aggregating the prediction score of all the three classifiers as discussed in the above sections. The prediction or confidence score of each classifier got aggregated by using a soft voting approach. In general, there are the two common approaches for data aggregation viz., soft voting approach, and hard voting approach. Soft voting approach uses the concept of prediction probabilities, wherein output is received in terms of probabilities. In case of multiple classifiers, final output is obtained by taking the average of the probabilities of each of the class produced by individual classifiers and whichever class gets the highest average is considered as the final prediction. Whereas, in hard voting approach, the output of each individual classifier is obtained in terms of binary value (for e.g.: 0/1) and finally the output class, which was returned in majority by the classifiers is selected. In general, the soft voting approach is preferred over hard voting approach as soft voting approach provides more information in terms of probability. Therefore, in this paper, we adopted soft voting approach for combining the classifier's prediction scores (Saha and Ekbal, 2013).

Soft voting-based classifier assigned a label to the test data by adding the overall confidence score of all classifiers corresponding to a particular class (Kima et al., 2011). Maximum confidence score for a particular class was decided based on aggregation of the confidence score of all the three classifiers. The aggregation of

the classifier's output with the soft voting approach can be mathematically expressed as:

$$C = \arg\max \sum_{j=1}^{3} W_j * P_i^j \tag{24}$$

$$C = \left\{ \begin{matrix} 1 & \text{if } p_{iR} > P_{iNR} \\ 0 & \text{otherwise} \end{matrix} \right\} \tag{25}$$

where, C is a final decision of classifier on the test data. If the classifier output is 1, it indicates assigned label to be road region, while the classifier output 0 indicates the assigned label to be the non-road region. The aggregated confidence score for road and the non-road region is indicated as P_{iR} andP_{iNR}.

2.5.1. Confidence score (CS) generation

The soft voting aggregation approach needs the confidence score (CS) of each classifier. The nearest neighbor classifier classifies road and non-road region based on the distance measured from a particular class. In the proposed system, NN classifier measure the distance between test points and input feature vectors and stores them as a distance vector D. Further, the measured distance, which is stored in terms of distance vector D is utilized to generate confidence score for each classifier. Let's assume the distance vector D to be (d1, d2 ... d6). Then, confidence score can be mathematically expressed as:

$$P = \left\{ 1 - \frac{d_i}{\sum_{i=1}^{6} d_i} \right\} \tag{26}$$

where, *P* represents generated confidence score, *d* represents measured distance of each classifier from a particular class.

3. Results and discussion

This section presents the visual and quantitative results of the proposed method. The proposed method was implemented on Matlab 2017a software using a personal computer (PC) with Intel Core i7 processor having speed of a 3.7 GHz and internal memory of 4 GB. The total 4000 sample images were used to build the classifier. The dataset was divided into the ratio of 75% and 25% for training and testing the classifier. Prior to developing the classifier, the input images of unstructured road were categorized into road and non-road regions. After segmentation, the chromatic color feature information such as *a, *b, and Hue was extracted to train the classifier. However, various pre-processing algorithms such as image down sampling, image smoothening, and image sharpening were applied before extracting the feature information to enhance the image quality through suppression of redundant pixels by applying a series of appropriate filters as elaborated in section 2. The prime objective of pre-processing was to reduce the computational burden on hardware machine in addition to the improvement in pixel contrast of the relevant features. Subsequently, independent chromatic feature set corresponding to each color image were formed and applied as an input to build three independent NN classifiers. Furthermore, the trained classifiers were used to classify a test images into road and non-road regions using a soft voting aggregation approach. The proposed system assigned a label 1, if the test image belongs to the road regions, else 0 if the test image belongs to non-road region based on the value of overall confidence score aggregated by the soft voting approach. The visual results yielded by the proposed system are shown in Fig. 5, where the black pixel in output image shows the non-road region and the white pixels shows the road region. The quantitative results in terms of confidence score and binary output as "road region" and "non-road region" for three test sample images is provided in Table 1. As evident in Table 1, all the three classifiers yields the output as a road and non road region based on the distance measured between test feature vector and training feature vector. The calculated distance reflects the closeness of the test sample image to a particular class (i.e. road or non-road region). Results also demonstrate that the hard voting approach takes the final decision mere on the basis of output response (i.e. binary value 0 or 1) of all the three classifiers and does not take into account the confidence score (i.e. prediction score in terms of probability) alike soft voting approach.

3.1. The metrics used for performance evaluation

The performance of the proposed road detector system was finally evaluated in terms of the metrics such as precision, recall, intersection over union (IOU), true positive rate (TPR), and false positive rate (FPR).

The recall of the system gives information about the fraction of points predicted by the system correctly as a road and non-road regions. Whereas, precision denotes the fraction of the positive class correctly predicted by the system (Cheng et al., 2009). The ground truth of the road and the non-road region is essential information for calculating the precision and recall of the road detection system. Therefore, ground truth in the form of the road and non-road regions was manually labeled in all the test images. The precision and recall performance measures can be expressed as follows:

$$\text{Precision} = \frac{\text{Matched}(S, R)}{|R|} \tag{27}$$

$$\text{Recall} = \frac{\text{Matched}(R, S)}{|S|} \tag{28}$$

where *R* indicates the ground truth of the test image and *S* denotes the output image of the corresponding classifier indicating detected road and non-road region in the test image.

Intersection over Union (IOU)

IOU measurements quantify the percentage of overlap between the segmented output and labeled ground truth. Mathematically, IOU metric can be expressed using Eq. (29).

$$\text{IoU} = \frac{\text{Ground Truth} \cap \text{Prediction}}{\text{Ground Truth} \cup \text{Prediction}} \tag{29}$$

True Positives (TP): The system predicted road region as road is known as true positive prediction.

True Negatives (TN): The system predicted non-road region as non-road region is known as true negative predictions.

False Positives (FP): The system predicted non-road region as road region is known as false positive prediction.

False Negatives (FN): The system predicted road region as non-road region is known as a false-negative prediction.

True Positive Rate (TPR)

TPR gives information about the actual road candidates, which are correctly detected by the system. Mathematical expression for TPR calculation is shown in Eqs. (30).

$$\text{TPR}\% = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100 \tag{30}$$

False Positive Rate (FPR)

FPR is defined as the fraction of non-road region candidates, which gets identified as road region by the system. Mathematically, it can be expressed using Eq. (31) as follows:

$$\text{FPR}\% = \frac{\text{FP}}{\text{FP} + \text{TN}} \times 100 \tag{31}$$

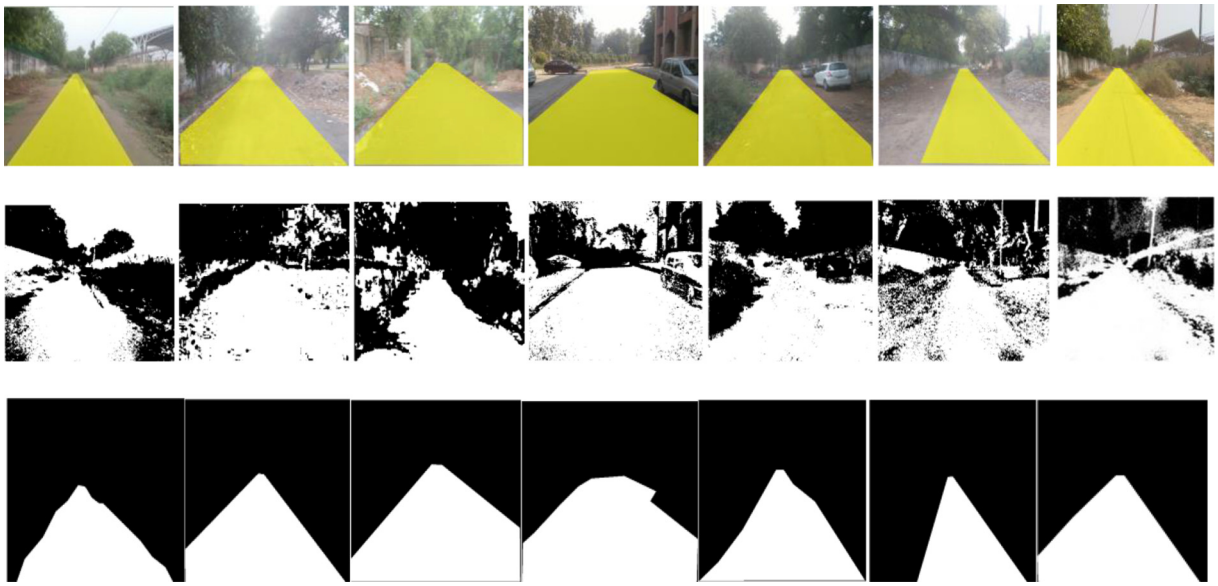

Fig. 5. - Sample images of detected road region with proposed classier: Row 1: Sample images of ground truth; Row 2: Detection of road region using hard voting approach; Row 3: Detection of road region using soft voting approach.

Table 1
Decision-making process of proposed system.

Test Sample No.	Distance measured	Classifiers output	Hard voting based output	Confidence Score (Road, Non-road)	Overall Confidence score (Road, Non-road)	Soft voting based output
1.	100	Road	Road	(0.902, 0.098)	(2.7236,3.72)	Non-road
	200			(0.196,0.803)		
	50	Road		(0.950,0.049)		
	70			(0.0686,0.931)		
	500	Non-road		(0.509,0.490)		
	100			(0.098,0.901)		
2.	50	Road	Road	(0.9219, 0.0781)	(2.9064, 3.0938)	Non-road
	150			(0.2344, 0.7656)		
	100	Road		(0.8438,0.1563)		
	120			(0.1875, 0.8125)		
	200	Non-road		(0.6875, 0.3125)		
	20			(0.0313, 0.9688)		
3.	200	Non-road	Non Road	(0.7849,0.2151)	(2.9246, 3.0754)	Non-road
	100			(0.1075,0.8925)		
	100	Non-road		(0.8925,0.1075)		
	80			(0.0860, 0.9140)		
	200	Road		(0.7849, 0.2151)		
	250			(0.2688, 0.7312)		

3.2. Simulation results

The classification results in terms of Precision and Recall performance metrics is presented in Table 2. The numerical results given in Table 2 demonstrate that the proposed system obtained the mean precision score of 0.9679 at the recall rate of 0.9692, which seems to be quite satisfactory for unstructured road detection tasks.

Table 2
Performance comparison between soft voting and hard voting based NN classifiers in term of precision and recall score.

Performance measures	NN Classifier with Hard Voting approach	NN Classifier with Soft Voting approach
Precision	0.6040	0.9679
Recall	0.9330	0.9692

As stated earlier, in order to evaluate the performance of the system, we applied both kind of voting approaches (i.e. soft voting and hard voting) with the ultimate aim of aggregating the output responses of all the classifiers. However, classifier based on the soft voting approach yielded the better visual results in comparison to the classifier based on hard voting approach as illustrated in Fig. 5.

Fig. 6 presents precision vs. recall (P-R) curve for both soft and hard voting approaches. A P-R curve shows the tradeoff between precision and recall for different thresholds. A large area covered under the curve represents the value of both metrics (i.e. precision, and recall) to be high, where high precision relates to a low false positive rate, and high recall relates to a low false negative rate. Precision is also known as the Positive Predictive Value (PPV) and Recall is also known as True Positive Rate (TPR). Therefore, high value of both, precisions and recall ensure the good performance of the classifier.

From the P-R plot curve, we can see that soft voting and hard voting approaches achieved the mean precision score of 0.9679 and 0.604 at recall rate of 0.9692 and 0.933, respectively. The

precision and recall curve were also plotted individually for different number of images as illustrated in Fig. 7 (a) and 7(b), which also proves the efficacy of soft voting approach over hard voting approach.

3.3. The performance evaluation

3.3.1. Comparison between the individual classifiers and the proposed method

The results of the proposed method were compared with the output of the individual classifiers, which were trained with separate chromatic features information such as *a, *b and Hue, in terms of TPR, FPR, IOU and detection accuracy as demonstrated in Table 3. The individual classifiers (i.e., C_{NN1}, C_{NN2}, C_{NN3}) achieved the detection accuracy of 60.83, 61.09, and 63.79, respectively, while the proposed method achieved detection accuracy of 97.89 % that is far better than the detection accuracy achieved by the classifier trained using individual chromatic feature information and hard voting aggregation based approach. The results presented in Table 3 also demonstrate the superior results in terms of TPR, FPR, and IOU compared to other classifiers.

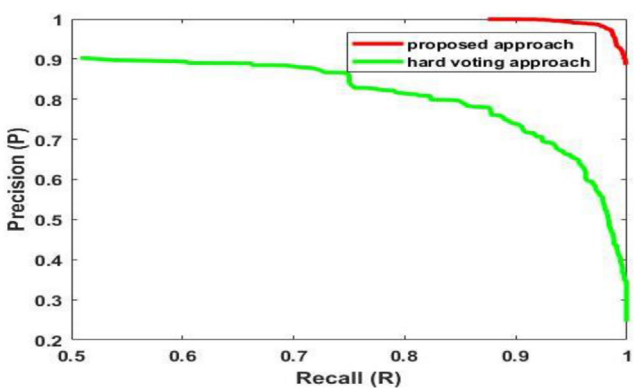

Fig. 6. Precision and recall plot of proposed system based on soft voting and hard voting system.

3.3.2. Compared with current state of art methods

Currently, CNN and deep learning based methods are considered to be one of the most popular methods for developing vision-based detection systems. Hence, in this work, we also compared the performance of the proposed system with the CNN and other deep learning based approaches and a comparative study was performed in terms of prediction time, TPR, FPR, IOU, and detection accuracy in order to assess the effectiveness of the proposed method over the other current state-of-the-art methods. Table 4 presents the comparative results between proposed detection method and other methods such as KNN & SVM (Shang et al., 2013), CNN (Pan et al., 2017), Vanishing point (Li et al., 2018), and Deep Lab V3+ (Baheti et al., 2020) in terms of most prominent performance matrices viz., prediction time, TPR, FPR, IOU, and detection accuracy. For the fair comparison among the results, we tested the different classifiers including the proposed method using the same data set, and hardware. From the results shown in Table 4, it is clear that the proposed system achieved high accuracy, TPR, IOU and requires less processing time with low FPR in comparison to other state of art methods reported in the literature. The proposed road detection method obtained the detection accuracy of 97.89%, IOU of 96.08 %, and TPR of 96%, which are comparatively higher than those of other common artificial intelligence-based road detection methods such as vanishing point-based road detection (Li et al., 2018), KNN (Shang et al., 2013), SVM (Shang et al., 2013), CNN (Pan et al., 2017), and semantic segmentation (Baheti et al., 2020). In addition to the above advantages, the proposed system took the less average processing time in comparison to the other state of art methods as illustrated in Table 4.

We also compared the performance of the proposed system with other deep learning based algorithms in terms of precision, and recall as illustrated in Table 5. The results depicted in Table 5 demonstrate that proposed method achieved the highest accuracy

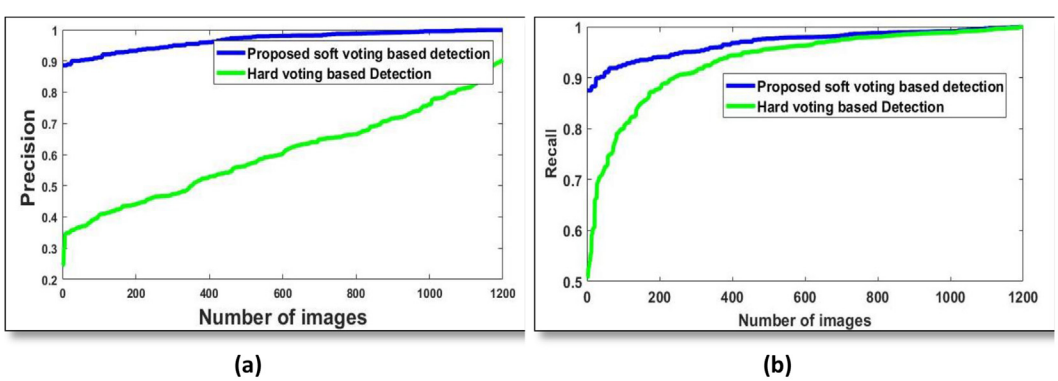

Fig. 7. Individual representation of precision and recall plot for varying number of images using proposed approach based on soft voting and hard voting approach; (a) Precision plot, (b) Recall plot.

Table 3
The Performance comparison among the individual classifiers and the proposed method.

S. No	Machine Learning Models	TPR (%)	FPR (%)	IOU (%)	Accuracy (%)
1	Classifier-1 (C_{NN1}) trained with chromatic feature *a	61.03	30.02	61.43	60.83
2	Classifier-2 (C_{NN2}) trained with chromatic feature *b	60.98	30.23	62.09	61.09
3	Classifier-3 (C_{NN3}) trained with chromatic feature Hue	62.67	29.56	62.78	63.79
7	Classifier with combined features & hard voting approach	89.00	8.65	91.23	90.43
8	**Proposed method**	**96.90**	**2.23**	**96.08**	**97.89**

Table 4
Performance comparison of different methods for the classification of road and non-road regions.

S. No.	Road detection Algorithms	TPR (%)	FPR (%)	IOU (%)	Accuracy	Average Testing time (sec)
1	KNN (Shang et al., 2013)	89.76	13.08	89.02	90.43	1.00
2	SVM (Shang et al., 2013)	90.80	10.65	91.06	93.46	1.50
3	CNN (Pan et al., 2017)	95.07	9.72	94.26	96.12	1.78
4	Vanishing point (Li et al., 2018)	93.00	11.40	93.32	94.21	1.3
5	DeepLabV3 + (Baheti et al., 2020)	81.58	16.72	80.85	81.05	1.53
6	**Proposed method**	**96.00**	**2.23**	**96.08**	**97.89**	**0.49**

Table 5
Quantitative comparison of proposed method & deep learning based approaches.

S. No.	Algorithms	Precision (%)	Recall (%)	Accuracy (%)
1	CNN (Pan et al., 2017)	97.23	93.6	96.12
2	DeepLabV3+(Bhakti Baheti et al,0.2020)	83.54	79.01	81.05
3	**Proposed method**	**96.79**	**96.92**	**97.89**

and recall score of 97.89 and 96.92, while results obtained in terms of precision were comparable to that of CNN but higher than that of DeepLab V3 + method. However, the processing time taken by the CNN was very high compared to the proposed method as well as other conventional machine learning based methods. From the results, we can infer that the overall performance of the proposed system is very promising and hence provides a great hope to provide the potential solution for autonomous vehicles for detection of unstructured roads in rural and urban areas of the developing country like India.

4. Conclusion

In this paper, we developed an unstructured road detection system based on NN classifier and soft voting approach to detect and classify the road and non-road regions. First of all, we built three nearest neighbor classifiers with *a,*b, and H chromatic information and thereafter aggregated their output based on soft voting approach for determining the final output response as road and non-road regions. The soft voting approach was utilized for aggregation of classifier's output based on their confidence score. We evaluated the performance of the proposed road detection system in terms of performance metrics such as precision, recall, accuracy, and training time. A total of 4000 images were captured under various light conditions for evaluating the system performance. The proposed system obtained the mean precision of 0.9679 at the recall rate of 0.9692 with the detection accuracy of 97.8%, respectively. We compared the performance of our system with those of existing state of art methods such as vanishing point algorithm, SVM, KNN, CNN, and DeepLabV3 + algorithms. From the experimental results, we found that the soft voting based proposed method is more competent and powerful for the classification of unstructured roads in comparison to the other state of art methods. Hence, authors recommend its use for the specific application of unstructured road detection and its classification as a road and non-road regions thereby. However, despite of achieving good classification accuracy for the above stated problem, the proposed system may not perform well in the presence of other moving objects such as car, trucks, bus, pedestrian, etc. on road that might be considered as a future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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